

Forecasting Timber Prices in Volatile Markets: An Evaluation of Model Complexity and Temporal Frequency

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Abstract

Accurate timber price forecasting is crucial for effective forest management and a strategic response to changing commodity markets. This study quantifies the forecasting accuracy of univariate statistical and machine-learning methods at weekly, monthly, and quarterly time intervals using price data on Polish pine sawlogs and pulpwood. We evaluate Autoregressive Integrated Moving Average (ARIMA), Exponential Smoothing (ETS), Dynamic Harmonic Regression (DHR), a Light Gradient Boosting Machine (LightGBM), and naive baselines. A major practical issue discussed is whether temporally aggregating weekly forecasts yields better accuracy than direct monthly modeling. Theoretical analysis shows that direct monthly modeling is almost always better than upward aggregation, since higher-frequency data is statistically noisier, which leads to degradation of predictive accuracy at lower frequencies. In a purely univariate context, statistical models perform better than the LightGBM model, which indicates that, in order to be able to reveal meaningful signals, machine learning models need access to exogenous macroeconomic variables. Furthermore, we quantify model resilience during the COVID-19 structural market break. While all models failed to predict the unprecedented price surge, the simple naive baseline adapted to newly established price levels much faster than parametric models retaining outdated pre-shock distributions. Ultimately, for long-term strategic planning, direct monthly modeling using ARIMA or a naive baseline remains the most reliable and efficient approach for forestry organizations.

Keywords: Timber Price Forecasting, Forest Management, Multi-Horizon Forecasting, Polish Timber Market, Timber Price Shocks, Temporal Aggregation, Structural Breaks

1 Introduction

Precise timber price prediction is a key component for proper forest management. It helps the forest owners not only with harvest planning but also with adjusting the strategy in a changing world commodity market (Leskinen and Kangas, 1998, 2001). In Poland, this is even more complicated, as the majority of the forest market is dominated by the State Forests National Forest Holding, which largely determines pricing policies and does not reflect European market conditions (Górna et al., 2022; Wysocki et al., 2025a). While historical Polish market data exhibits systematic variations suitable for short-term trend prediction (Adamowicz and Górna, 2020), the timber market remains vulnerable to external macroeconomic anomalies and structural supply chain shifts (Matasová, 2025).

In recent years, the timber price forecasting literature has seen an ongoing debate regarding the level of model complexity (Ma et al., 2024). Traditionally, univariate time series models such as Autoregressive Integrated Moving Average (ARIMA) and Exponential Smoothing (ETS) have remained the standard tools for short-term predictions. However, an increasing number of studies advocate for the use of machine learning (ML) architectures, based on their ability to capture non-linear price dynamics (Verly Lopes et al., 2021). Yet, the efficacy of complex ML models in strictly univariate environments, without the exogenous predictors like construction indices or substitute product prices remains heavily contested.

Temporal resolution has been underexamined in forestry economics despite its influence on forecasting precision. Even though most studies fix price data to a single frequency, they don't assess how varying data

^{*}Funding: This research was supported by the National Science Center in Poland (grant no. 2021/43/B/HS4/03371).

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intervals affect model outcomes. At least in theory, forecasting high-frequency values and then summarizing them may outperform direct low-frequency projections. But structural shifts during the pandemic revealed how even sophisticated models fail under extreme market disruptions. The volatility introduced by the COVID-19 pandemic exposed weaknesses in both traditional and machine-learning-based approaches.

To fill closely related gaps, this paper compares different forecasting methods over time for timber prices in Polish auctions at multiple time frequencies. Polish timber auctions are a major source of forest market data. Two main assortments of timber, pine sawlogs and pine pulpwood, are considered in this paper. A wide range of univariate models are compared in this study, including naïve baselines ETS, ARIMA, Dynamic Harmonic Regression (DHR), as well as a gradient-boosted tree algorithm (LightGBM). Using walk-forward cross-validation for weekly, monthly, and quarterly time steps separately, this study examines how model complexity, time aggregation, and robustness to structural shocks interact.

Based on the synthesis of the current literature and the empirical objectives of this study, we propose the following research hypotheses:

- **Hypothesis 1:** In a strictly univariate forecasting setting, statistical models outperform complex machine learning algorithms in predicting timber prices.
- **Hypothesis 2:** Direct forecasting at lower temporal resolutions (e.g., monthly) yields higher predictive accuracy than the temporal aggregation of higher-frequency (e.g., weekly) forecasts.
- **Hypothesis 3:** Simple naïve models recover from unprecedented structural market breaks more rapidly than parametric or seasonal models.

Hypothesis 1 emerges from the ongoing discussion about model complexity. Studies indicate a growing reliance on artificial neural networks and machine learning approaches to model price dynamics, primarily due to their ability to represent non-linear patterns (Verly Lopes et al., 2021; Ma et al., 2024). Evidence suggests that multi-layer perceptrons and hybrid architectures outperform standard ARIMA models in forecasting price reversals and equilibrium levels (Kozuch et al., 2023; Lamichhane et al., 2023). Yet, for now, these models generally fail to outperform classical statistical methods unless they incorporate external variables such as macroeconomic indicators, spatial information, or multiple data streams (Mei et al. (2010); Thao and Ly Duc (2025); Wysocki et al. (2025b)). In cases involving only single-variable price series, complex machine learning systems struggle to detect meaningful signals without additional inputs like construction activity indices, substitute product prices, or broader housing market demand trends (Parajuli et al., 2016; Banaś and Utnik-Banaś, 2021).

Hypothesis 2 targets a key limitation in hierarchical forecasting within forestry. Most studies assess timber prices at just one frequency - monthly (Malaty et al., 2007; Michinaka et al., 2016) or quarterly (Banaś and Utnik-Banaś, 2021; Matasová, 2025) - without comparing model performance across time scales. Contrary to trends in other forecasting fields, we propose that higher-frequency data, mainly when heavily imputed, adds noise that hurts the accuracy of lower-frequency predictions. As a result, monthly modeling is expected to yield better statistical results than aggregated weekly forecasts, which suffer from increased variability due to imputation errors.

Hypothesis 3 stems from the fact that model resilience under macroeconomic anomalies remains a key concern. Structural market shifts, such as pandemics, natural disasters, and geopolitical crises, challenge even the most set up modeling systems (Hernandez-Diaz et al., 2024). A time-series-only approach fails to account for such disruptions, as shown by Malaty et al. (2007), and classical or advanced models produced inaccurate forecasts during the COVID-19-driven lumber price surge (Verly Lopes et al., 2021; Lamichhane et al., 2023). The naïve model adjusts immediately to new price levels after a structural break. At the same time, parametric and seasonal models retain outdated pre-shock distributions. This rapid adaptation allows the naïve model to recover faster from abrupt changes than alternative approaches.

The remainder of the paper is structured as follows. In Section 2, we review the existing literature on timber price forecasting and model complexity. Section 3 discusses the data, including temporal aggregation methods and outcomes of stationarity tests. Section 4 sets forth the empirical approach and lays out the details of statistical and machine learning models, an expanding-window cross-validation framework, and a cross-frequency alignment strategy. In Section 5, the results of the forecasting exercise are revealed, the impact of COVID-19 as a structural break is illustrated, and the practical implications of cross-frequency comparisons are theoretically elucidated. Finally, in Section 6, we conclude the study.

2 Literature review

Forecasting timber prices accurately is crucial for optimizing forest planning, scheduling harvests, and being able to handle the changes in global commodity markets (Leskinen and Kangas, 1998, 2001). In Central Europe, regional timber prices are strongly connected, responding uniformly to macroeconomic shifts despite differences in local absolute prices (Gejdoš and Potkány, 2017; Gejdoš et al., 2019; Wysocki et al., 2025a). Nevertheless, the Polish timber market poses multiple challenges for time series modeling because of the almost-monopoly of the State Forests National Forest Holding (*PGL LP*), which heavily controls pricing policies and distorts local market structure (Górna et al., 2022; Wysocki et al., 2025b). Although previous studies in Poland have shown that timber prices exhibit systematic variability that can be exploited for short-term trend prediction (Adamowicz and Górna, 2020), ultimately, external factors and structural changes in the supply chain govern local price determination (Matasová, 2025).

Historically, univariate time series models, especially Autoregressive Integrated Moving Average (ARIMA) and Exponential Smoothing (ETS), have been the main tools for short-term timber price forecasting [Ma et al. \(2024\)](#). The first quantitative attempts are recorded in [Buongiorno and Balsiger \(1977\)](#), where the authors employed Box-Jenkins ARIMA models for French timber products and explained the superiority of these models over simple extrapolation methods, while also pointing out that some assortments essentially behaved as random walks. Continuing these efforts, [Yin \(1999\)](#) showed that although ARIMA models were quite accurate for one-quarter-ahead forecasts in the US South, after differencing, the price series of several wood products appeared to be white noise, implying that the only valid predictor was a random walk. In Polish forestry, classical models were the most straightforward way to forecast industrial biomass and timber assortments. The authors are of the opinion that ARIMA is still the most reliable model for highly volatile assortments, while the Winters-Holt (ETS) has been very efficient for stable products ([Górna et al., 2022, 2023](#)). This product-specific adaptability aligns directly with our current findings, where ARIMA achieved the highest accuracy for monthly standard pine sawlogs and ETS excelled for weekly pine pulpwood.

The issue of whether timber prices follow a random walk or revert to the mean is a recurring theoretical debate. Based on a rational expectations equilibrium model, [McGough et al. \(2004\)](#) showed that even an efficient market can produce stationary prices that are well represented by an $AR(1)$ process. They also forecast right-skewed price distributions with occasional sharp spikes theoretically - something that exactly fits the extreme right-skewness seen in our cross-validation error distributions. On the other hand, [Khajuria et al. \(2012\)](#) estimated Canadian timber prices as a mean-reverting process with a stochastic trend and realized that traditional unit root tests can fail to uncover stationary components. As per our combined ADF and KPSS stationarity analysis, all the series under study become stationary after first differencing, and weekly pulpwood exhibits the least stationarity. Therefore, our multi-frequency analysis not only confirms the strength of the naïve model, which was very competitive and even came first for quarterly forecasts and monthly pulpwood, but also showed its effectiveness as a strong baseline in noisy, limited-sample environments.

The issue of seasonal changes in timber prices that can be utilized is yet another matter of dispute. On one hand, some authors support the use of seasonal models, such as SARIMA, capable of capturing repetitive changes ([Banaś and Utnik-Banaś, 2021](#); [Górna et al., 2022](#)), on the other hand, some have even based their sales operations on the optimization of seasonal peaks ([Marier et al., 2014](#)). However, the evidence from the Polish market is different. [Banaś and Kożuch \(2019\)](#) revealed that, while cyclic fluctuations and macroeconomic trends are present, their seasonal variations are outweighed, accounting for only around 20% of the variance in Polish timber prices. Our investigation strongly reaffirms their finding by showing that Seasonal naïve models, when evaluated across all product-frequency combinations, perform very poorly and consistently (MASE above 1.4 everywhere), implying that there is no exploitable annual seasonality at all.

Research has shown a trend towards using artificial neural networks (ANNs) and machine learning (ML) as the primary tools for capturing the non-linear movements of prices ([Ma et al., 2024](#); [Verly Lopes et al., 2021](#)). There is evidence that multi-layer perceptrons (MLPs) and hybrid networks can outperform traditional ARIMA models by a large margin in predicting price turning points and overall price levels ([Kożuch et al., 2023](#); [Lamichhane et al., 2023](#); [Thao and Ly Duc, 2025](#)). However, our study found that the gradient-boosted tree framework, LightGBM, consistently underperformed purpose-built statistical methods. This difference points to an important boundary condition - to surpass statistical baselines, machine learning models typically require exogenous macroeconomic indicators, spatial data, or multivariate structures ([Mei et al., 2010](#); [Thao and Ly Duc, 2025](#); [Wysocki et al., 2025b](#); [Zhou and Buongiorno, 2006](#); [Kolo and Tzanova, 2017](#)). When it comes to univariate data, it is difficult for complex ML models to find a new signal without the help of external factors like construction indices ([Banaś and Utnik-Banaś, 2021](#)), prices of substitute products ([Parajuli et al., 2016](#)), or macroeconomic housing demands ([Deb et al., 2023](#)).

One of the enduring challenges faced by modeling frameworks – including statistical and machine learning models – is their fundamental vulnerability to structural market shifts that have never been experienced before, such as natural disasters, geopolitical conflicts, and global pandemics ([Hernandez-Diaz et al., 2024](#); [Górna et al., 2023](#); [Sivaram, 2014](#)). A study by [Malaty et al. \(2007\)](#) indicated that, regardless of the level of model sophistication, a time-series-only approach is not sufficient to handle structural shifts. As a result, to capture extreme price spikes, [Leskinen and Kangas \(1998\)](#) proposes modelling them as independent Bernoulli processes. Unfortunately, neither classical nor advanced LSTM models were able to provide precise forecasts of exceptional lumber price increases driven by the COVID-19 housing boom, as the training datasets lacked historical periods of high volatility ([Verly Lopes et al., 2021](#); [Lamichhane et al., 2023](#)). Our study directly measures this weakness and finds that the COVID-era price spike led to abnormally large forecast errors across all models, and degradation factors in MASE reached up to 8 times. In fact, our research adds to this debate by providing evidence that the naïve model bounces back from structural breaks much more quickly than either parametric or seasonal models since it instantly changes to the new price level, whereas the others still stay stuck with the pre-shock distributions, which are by then time- outdated.

Last but not least, the effect of temporal frequency on forecast quality is a fundamental issue that remains largely unaddressed. Oftentimes, research identifies timber prices at only one frequency e.g., monthly ([Buongiorno and Balsiger, 1977](#); [Malaty et al., 2007](#); [Michinaka et al., 2016](#); [Gejdoš et al., 2019](#))

or quarterly (Yin, 1999; Banaś and Utnik-Banaś, 2021; Banaś and Kożuch, 2019; Matasová, 2025; Parajuli et al., 2016; Kożuch et al., 2023) time points, this is done without the systematic comparison of the model performance at various temporal resolutions. Indeed, this paper is the first to be thorough when it comes to the cross-scale benchmarking of forecasting models - weekly, monthly, and quarterly horizons. Diebold-Mariano tests definitively indicate that direct monthly modeling statistically dominates the upward aggregation of fine-grained weekly forecasts, demonstrating that higher-frequency data, particularly when subjected to substantial imputation, introduces noise that degrades lower-frequency predictive accuracy.

3 Data

The dataset for this research comprises transactional timber prices from the e-Drewno Individual Auctions, the sales channel that most closely represents an open market. Even though the platform accounts for the smallest fraction of the total sales volume, it is the only Polish timber market without institutional entry barriers. The sales of the State Forests of Poland are divided among three channels: e-Drewno Individual Auctions, e-Drewno System Auctions, and Offer Sales Auctions (commonly referred to as *PL-D*). The latter two are wholesale channels, with the Offer Sales Auctions channel handling about 80% of the volume. Nevertheless, these wholesale channels pose several analytical challenges. Firstly, sales are infrequent, usually only twice a year. Secondly, very limited data is made public. And finally, the institutional constraints, along with the intricate auction rules, make it very challenging to model these prices. Studying these channels would require a completely different, general-equilibrium-style framework to model offer grading, purchase-history requirements, institutional minimum price, and multi-criteria (point-and-volume) auction outcomes. For a detailed description of the sales channels, see Wysocki et al. (2025a).

This study is based on two primary timber products, which are the most significant in terms of volume sold: the "w_standard" pine assortment (Pine Logs) and the "s_s2a" pine assortment (Pine Pulpwood). The prices were sourced from the Polish Economic Chamber of the Wood Industry¹. As depicted in Figures 1a and 1b, the logs series starts in January 2009 and the pulpwood series starts in March 2009, while both end in April 2025.

The first important issue is that the raw data points are irregularly spaced, because the auctions do not happen every day, not all regions offer the same assortment of wood at every auction, and some regions offer selected assortments only periodically. These issues might introduce spatial, temporal, and seasonal biases, but they are inherent features of commodity auction markets.

To aggregate the raw auction prices into time series, we gathered all available transnational prices from all regions and averaged them into daily observations using volume weighting. Our main objective is to estimate the average transaction price in the market, weighting each observation by its economic significance (i.e., transaction volume). While this method does not systematically address spatial effects, higher-volume transactions are likely to be more informative about market prices, whereas low-volume observations may reflect idiosyncratic or illiquid conditions, thus we decided to use the volume-weighted average price as our modelling target.

To construct the time series, we adopted the last observation carried forward (LOCF) methodology. The complete daily series were generated by re-indexing the existing data points onto a full calendar and filling the missing observations (i.e., dates without an auction) by propagating the most recent available price. This is the most natural choice, as between the auctions, the last available price should reflect prevailing market valuation. However, this method might introduce long periods of flat prices due to the absence of auctions. We addressed this by checking the sparsity of auctions and found no specific pattern indicating it would be an issue. The only such period we identified occurred around 2017 (see Figures 1a and 1b), but a single such case should not inflate the first-order autocorrelation by much.

The imputed daily series are aggregated to three analysis frequencies: weekly ($f = W$), monthly ($f = M$), and quarterly ($f = Q$). All frequencies use end-of-period alignment. At each frequency, the aggregate value is defined as the arithmetic mean of daily prices within the respective period:

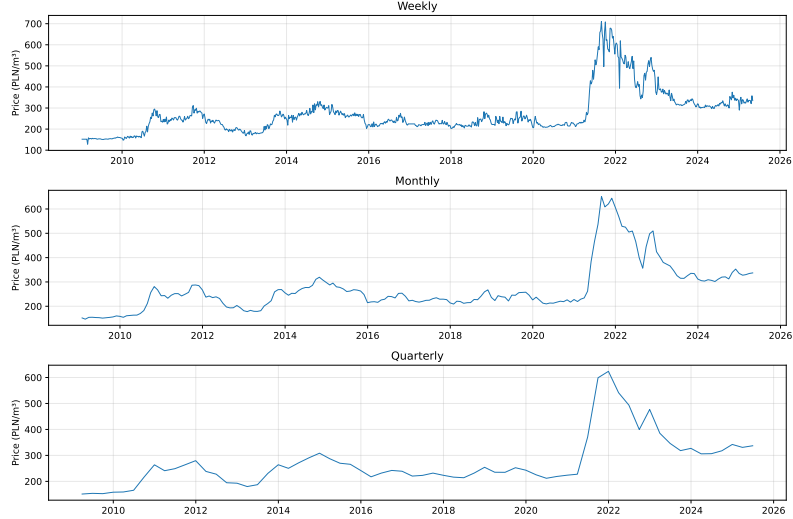
$$\bar{y}_t^{(f)} = \frac{1}{n_t^{(f)}} \sum_{d \in \mathcal{D}_t^{(f)}} y_d \quad (1)$$

where $\mathcal{D}_t^{(f)}$ denotes the set of daily observations in period t at frequency f , and $n_t^{(f)} = |\mathcal{D}_t^{(f)}|$. The arithmetic mean is preferred over end-of-period sampling because it provides a more representative summary of price levels within each aggregation window, which is pertinent to timber market planning, where average procurement cost, rather than spot price, is the relevant economic quantity.

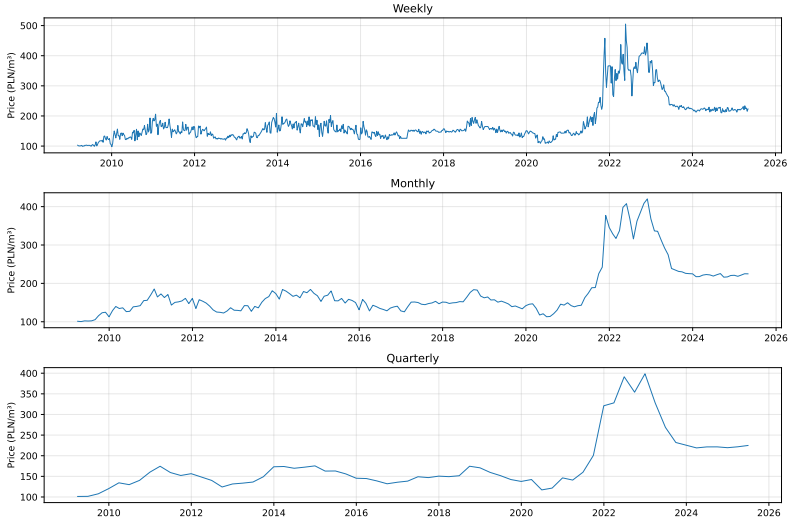
Table 1 summarizes the statistics of the Polish auction prices for pine sawlogs and pulpwood in the years 2009-2025. Pine sawlogs are typically more expensive, with average prices that are almost unchanged over time, from 272.09 (monthly) to 272.70 (quarterly). Pine pulpwood has average prices at a lower level, ranging from 176.83 to 177.14, depending on the frequency. Both assortments show very high price fluctuations in the mean price. The coefficient of variation (CV) is nearly the same across both products and across the different aggregation levels, indicating that the proportional variance is stable.

¹See: *Polska Izba Gospodarcza Przemysłu Drzewnego*, [PIGPD](#)

Figure 1: Prices of timber products in weekly, monthly, and quarterly resolutions.



(a) Pine Logs



(b) Pine Pulpwood

Note: Imputed daily series aggregated to three frequencies: Weekly, Monthly, and Quarterly. The series represent transnational prices from the e-Drewno Individual Auctions.

Table 1: Summary statistics of pine sawlogs and pulpwood prices from 2009 to 2025.

	Mean	Min	Max	St.D.	CV	Cnt
Pine Sawlogs (W)	272.24	128.00	710.09	99.98	0.37	852
Pine Sawlogs (M)	272.09	147.00	651.52	99.03	0.36	196
Pine Sawlogs (Q)	272.70	151.22	623.74	97.85	0.36	66
Pine Pulpwood (W)	177.13	97.89	505.16	67.73	0.38	844
Pine Pulpwood (M)	177.14	100.42	420.15	66.83	0.38	194
Pine Pulpwood (Q)	176.83	101.28	398.65	66.15	0.37	66

Note: W, M, and Q represent Weekly, Monthly, and Quarterly frequencies, respectively. CV stands for the coefficient of variation. The discrepancy in the number of observations between the products is a data artifact, as the auctions for pine pulpwood in 2009 began in March.

Temporal aggregation clearly reduces price extremes, which is an important consideration for the studied forecasting models (cf. Figures 1a and 1b). For instance, the highest price of pine sawlogs reduces from 710.09 at the weekly level to 623.74 at the quarterly level. At the same time, the lowest price increases from 128.00 to 151.22 (weekly vs. quarterly). There is also a fall in the extreme values of pulpwood.

Absolute dispersion decreases when we use temporal aggregation. Standard deviation for sawlogs changes from 99.98 (weekly) to 97.85 (quarterly), and for pulpwood, from 67.73 (weekly) to 66.15 (quarterly).

3.1 Stationarity Tests

To determine the appropriate order of integration and validate the model specifications, we apply two complementary tests to each product–frequency combination in levels, first differences, and log-levels. First, we used the Augmented Dickey-Fuller test (Dickey and Fuller, 1979) with automatic lag selection assuming regression with a constant and trend. Second, we utilize the Kwiatkowski–Phillips–Schmidt–Shin (KPSS) test (Kwiatkowski et al., 1992) under the same specification.

Table 2: Results of the unit root statistical tests.

Product	Frequency	Transformation	ADF Stat	KPSS Stat
Pine logs	W	levels	−2.712*	0.178**
Pine logs	W	diff	−5.621***	0.037*
Pine logs	W	log levels	−2.525	0.173**
Pine logs	M	levels	−2.614*	0.093*
Pine logs	M	diff	−9.022***	0.038*
Pine logs	M	log levels	−2.580*	0.090*
Pine logs	Q	levels	−2.789*	0.073*
Pine logs	Q	diff	−5.738***	0.037*
Pine logs	Q	log levels	−2.701*	0.070*
Pine pulpwood	W	levels	−1.709	0.292***
Pine pulpwood	W	diff	−6.048***	0.043*
Pine pulpwood	W	log levels	−1.892	0.303***
Pine pulpwood	M	levels	−2.468	0.142*
Pine pulpwood	M	diff	−4.287***	0.049**
Pine pulpwood	M	log levels	−2.328	0.147**
Pine pulpwood	Q	levels	−1.355	0.104*
Pine pulpwood	Q	diff	−5.069***	0.043*
Pine pulpwood	Q	log levels	−1.410	0.106*

Note: For the ADF test, the null hypothesis states that the time series contains a unit root and is therefore non-stationary, whereas the alternative hypothesis asserts that the series is stationary. The KPSS test reverses these assumptions, testing a null hypothesis of stationarity. Statistical significance at the 10%, 5%, and 1% levels is denoted by *, **, and ***, respectively.

We use the ADF and KPSS tests jointly, as in standard econometric practice, to determine the order of integration. In a case, especially when the ADF test does not have enough power to reject the null hypothesis of a unit root, whereas the KPSS test does reject the null of stationarity, then we may infer the series is non-stationary. Our weekly-level data exhibits this non-stationary pattern consistently as presented in Table 2. The test results at lower frequencies, however, are not as clear-cut. Upon differencing the data, both tests simultaneously indicate stationarity (i.e., the ADF test rejects the null, whereas the KPSS test does not). From this, the series are said to be integrated of order one, $I(1)$, which is compatible with choosing $d = 1$ in the ARIMA and DHR models. Lastly, the log-transformed series yields almost the same test outcomes as the raw data, showing that the transformation stabilizes the variance without changing the underlying integration order.

4 Methodology

We compare five classes of forecasting models, along with the seasonal naïve benchmark, spanning classical statistical, semi-parametric, and machine learning approaches. Each model is fitted to log prices, with automatic back-transformation to level prices for evaluation. No exogenous variables are used in any model – the comparison is strictly univariate.

4.1 Forecasting Models

All forecasting models in this study are fitted on log-transformed prices for two main reasons. First, the logarithmic transformation stabilises the variance of the series. Timber prices exhibit heteroscedasticity, with larger absolute fluctuations during high-price regimes (e.g., the 2021 price spike). Converting multiplicative dynamics on levels into additive dynamics on the logarithmic scale. Second, working in log space

ensures that additive error specifications in exponential smoothing and ARIMA models correspond to multiplicative error structures on the level, which is the natural formulation for strictly positive commodity prices. Formally, we define $z_t = \ln(y_t)$ and fit all models on $\{z_t\}$. Point forecasts $\hat{z}_{T+h|T}$ are back-transformed to level prices via $\hat{y}_{T+h|T} = \exp(\hat{z}_{T+h|T})$. To maintain interpretability across the expanding window cross-validation, final forecast evaluation metrics are reported on the original level prices.

4.1.1 Naïve Model

The naïve forecast sets the predicted value for all future horizons equal to the last observed value:

$$\hat{y}_{T+h|T} = y_T, \quad h = 1, 2, \dots, H \quad (2)$$

Under the random walk hypothesis $y_t = y_{t-1} + \varepsilon_t$, this is the minimum mean squared error predictor. Following, [Hyndman and Athanasopoulos \(2021\)](#) we include naïve model as a baseline against which the models can be evaluated.

4.1.2 Seasonal Naïve Model

The seasonal naïve forecast follows the same idea as its non-seasonal equivalent, but repeats the last observed seasonal cycle:

$$\hat{y}_{T+h|T} = y_{T+h-m\lceil h/m \rceil}, \quad h = 1, 2, \dots, H \quad (3)$$

where m is the seasonal period (52 for weekly, 12 for monthly, 4 for quarterly). This benchmark model allows us to assess whether predictable seasonal patterns exist in the data. It extracts the last m observations and tiles them forward cyclically over the forecast horizon.

4.1.3 Exponential Smoothing

The ETS framework (e.g., [Hyndman et al., 2008](#)) describes a set of models in state-space form, defined by the components: error (E), trend (T), and seasonality (S). We perform automatic selection of the model from six configurations that vary in trend (additive, no trend, damped or not) and seasonality. All models are applied to log-prices, so the additive parts in the logarithms correspond to multiplicative parts in levels. The length of the seasonal cycle is $m \in \{52, 12, 4\}$ depending on the frequency. The choosing of the model is based on using the corrected Akaike Information Criterion (AICc): $\text{AICc} = \text{AIC} + \frac{2k(k+1)}{n-k-1}$, where k is the number of parameters estimated and n is the sample size. AICc is chosen over AIC for the smaller monthly and quarterly series, where the finite-sample-size adjustment is particularly important.

4.1.4 Autoregressive Integrated Moving Average

For monthly and quarterly data, we employ the autoregressive integrated moving average model also allowing for the seasonality components, $\text{SARIMA}(p, d, q)(P, D, Q)_m$:

$$\phi_p(B) \Phi_P(B^m) (1 - B)^d (1 - B^m)^D z_t = \theta_q(B) \Theta_Q(B^m) \varepsilon_t \quad (4)$$

where B is the backshift operator. The non-seasonal autoregressive and moving average polynomials are defined as

$$\phi_p(B) = 1 - \phi_1 B - \dots - \phi_p B^p, \quad \theta_q(B) = 1 + \theta_1 B + \dots + \theta_q B^q,$$

and the seasonal polynomials are given by

$$\Phi_P(B^m) = 1 - \Phi_1 B^m - \dots - \Phi_P B^{Pm}, \quad \Theta_Q(B^m) = 1 + \Theta_1 B^m + \dots + \Theta_Q B^{Qm}.$$

The innovations satisfy $\varepsilon_t \sim \text{WN}(0, \sigma^2)$. The model orders are selected automatically via the stepwise search algorithm of [Hyndman and Khandakar \(2008\)](#), using AICc as the model selection criterion.

Seasonal ARIMA is applied only to monthly ($m = 12$) and quarterly ($m = 4$) data. For weekly data, where the seasonal period is $m = 52$, the SARIMA specification becomes considerably less practical. Seasonal components enter at lag 52, which requires long time series for reliable estimation and often results in weakly identified parameters and numerical instability ([Hyndman and Athanasopoulos, 2021](#)). Furthermore, weekly seasonal patterns are typically smooth and are not well captured by low-order seasonal ARMA terms. Consequently, we instead employ the Dynamic Harmonic Regression approach described in Section 4.1.5, which offers a more flexible and stable representation of such seasonality.

4.1.5 Dynamic Harmonic Regression

For weekly data, seasonality is modelled using Fourier terms included as exogenous regressors within a non-seasonal ARIMA error structure, following the approach of [Hyndman and Athanasopoulos \(2021\)](#). The model takes the form:

$$z_t = \sum_{k=1}^K \left[\alpha_k \sin\left(\frac{2\pi kt}{P}\right) + \beta_k \cos\left(\frac{2\pi kt}{P}\right) \right] + \eta_t, \quad (5)$$

where $P = 365.25/7 \approx 52.18$ denotes the average annual period expressed in weeks, K is the number of Fourier pairs, and η_t follows a non-seasonal ARIMA(p, d, q) process. The Fourier terms provide a flexible representation of smooth seasonal variation at the annual frequency and its harmonics, while the ARIMA error component captures the remaining short-run dynamics.

The number of Fourier pairs K is selected from the set $\{2, 3, 4\}$ by estimating each specification and choosing the model that minimises AICc. The orders of the ARIMA error process are determined using the stepwise procedure of [Hyndman and Khandakar \(2008\)](#). Compared to a full SARIMA(p, d, q)(P, D, Q)₅₂ specification, this approach substantially reduces the number of parameters and avoids estimation issues associated with high seasonal lags, while retaining sufficient flexibility to capture annual seasonal behaviour in weekly data.

4.1.6 Gradient Boosting Model

We employ a gradient-boosted decision tree model (LightGBM) ([Ke et al., 2017](#)) as our machine learning benchmark, motivated by its strong performance in practical forecasting applications (see, e.g., [Lainder and Wolfinger, 2022](#)). Because our analysis relies solely on endogenous data, the feature space consists entirely of lags and rolling window statistics tailored to the sampling frequency:

- Weekly: Lags $\{1, 2, 8, 13, 26, 36, 44, 52\}$; rolling means (windows 4 and 13); rolling standard deviation (window 4).
- Monthly: Lags $\{1, 3, 6, 12\}$; rolling mean and standard deviation (window 3).
- Quarterly: Lags $\{1, 2, 4\}$; rolling mean (window 2).

Hyperparameters are dynamically tuned at each cross-validation fold via Bayesian optimisation ([Akiba et al., 2019](#)) to adapt to the expanding training set. The internal validation relies on a time-series split with an expanding window, utilizing an initial 70% of the training data. We conduct 60 trials per fold, minimizing the mean absolute error. The complete search space is detailed in Table 3.

Table 3: Optimized hyperparameters search space.

Parameter	Range
Number of estimators	[50, 500]
Maximum depth	[3, 8]
Number of leaves	[8, 64]
Learning rate	[0.03, 0.2]
Minimum data in leaf	[20, 100]
L1 regularisation (α)	[0.0, 1.0]
L2 regularisation (λ)	[0.0, 1.0]
Row subsampling ratio	[0.7, 1.0]

Note: All hyperparameters were tuned using Bayesian optimization techniques withing the Optuna framework [Akiba et al. \(2019\)](#).

Finally, multi-step-ahead predictions are generated iteratively via a recursive strategy, where each new one-step prediction is appended to the feature set until the full forecast horizon is reached.

4.2 Model Evaluation

Model comparison is conducted via walk-forward cross-validation, which respects the temporal ordering of the data and mimics a realistic forecasting scenario. The procedure is defined as follows:

1. The initial training window covers the period from the first observation to 31 December 2013 (approximately 5 years of data).
2. At each fold i , a model is fitted on the training set $\{y_1, \dots, y_{T_i}\}$ and produces H -step-ahead forecasts $\{\hat{y}_{T_i+1}, \dots, \hat{y}_{T_i+H}\}$.
3. The training window expands by a fixed step size s , advancing the end of the training set, and the process repeats.
4. The procedure terminates when insufficient test data remain to fill the full forecast horizon.

To maintain a consistent forecasting horizon of approximately six months across all data frequencies, we change the forecasting stepsize s and horizon H based on the resolution. In the case of weekly data, $H = 26$ and $s = 4$ (i.e., we move forward in one-month steps). For monthly data, $H = 6$ and $s = 1$, and for quarterly data, $H = 2$ and $s = 1$.

We utilize an expanding window rather than a fixed window to allow the models to learn from an increasingly long historical series. This approach is highly appropriate given the relatively short available

sample and the non-trivial structural changes (cf. the 2021 price spike in Figure 1a) that enter the training set over time.

To assess the forecast performance, we compute four standard metrics on back-transformed level prices, averaged across all cross-validation folds: Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), Mean Absolute Percentage Error (MAPE), and Mean Absolute Scaled Error (MASE).

We utilize MASE (Hyndman and Koehler, 2006) as the main ranking metric due to its scale independence, robustness to near-zero actuals, and direct interpretability – values below 1.0 indicate improvement over a seasonal naïve benchmark. RMSE, MAE, and MAPE are reported alongside to provide supplementary scale-dependent and percentage-based perspectives on model accuracy.

4.3 Cross-Frequency Alignment Analysis

A key focus of this research paper is to determine whether finer-grained, temporally aggregated forecasts can outperform direct lower-frequency forecasts. To conduct a rigorous test, the 26-week forecasts from the weekly models are consolidated into calendar months using arithmetic means. These weekly-aggregated monthly forecasts, represented as $\hat{y}_t^{(W \rightarrow M)}$, are then aligned with the direct monthly forecasts, $\hat{y}_t^{(M)}$, by matching pairs of folds and months. Only the pairs with no fewer than 5 common observations are preserved for testing.

We further formalise the comparison by applying the Diebold–Mariano (DM) test Diebold and Mariano (1995) under the squared error loss function to test the null hypothesis of equal predictive accuracy. To appropriately measure overlapping multi-step-ahead forecast errors that are autocorrelated, the test statistic is computed using the Newey–West heteroscedasticity- and autocorrelation-consistent (HAC Newey and West, 1987) variance estimator with a bandwidth of $h - 1$, where h is the forecast horizon.

5 Results

The results presented in Table 4 suggest that forecasting performance depends heavily on temporal frequency and product type. For pine logs, dynamic harmonic regression is the only model clearly outperforming the naïve baseline at the weekly frequency (MASE = 0.704 versus 0.729). At the monthly level, ARIMA achieves the best absolute performance in the study (MASE = 0.693). On the other hand, weekly pine pulpwood prices seem to be best captured by exponential smoothing (MASE = 0.772), as smoother price movements with less structural change fit well the concept of adaptive level weighting. Nevertheless, when time granularity declines, the forecasting leverage of parametric tools weakens. The naïve random walk benchmark prevails in quarterly forecasts for both product types, as well as in the monthly pulpwood ones, because at such a low signal-to-noise ratio level, few - if any - exploitable patterns can be derived. Crucially, the seasonal naïve baseline is always the poorest model among alternatives (MASE > 1.4), indicating that isolated seasonal patterns in Polish timber prices are not strong enough to generate reliable forecasts. Lastly, the LightGBM algorithm does not emerge as the best in any case, at best attaining the runner-up position for quarterly data, thus highlighting the fundamental limitations of univariate tree-based machine learning methods against statistical methods specifically designed for this purpose when no exogenous regressors are available.

The time series of forecast errors presented in Appendix A indicate three distinct phases of performance. Up to 2020, forecast accuracy remains quite stable, even as the cross-validation windows grow larger. For most models, the MASE values range from 0.1 to 0.8. During this stable period, parametric models such as DHR and ARIMA perform slightly better than naïve baseline models, but the difference is small. The situation changes completely during the period from November 2020 to August 2021, when all models fail dramatically due to the unprecedented COVID-19 price spike. After 2022, the prediction errors are mostly back to the pre-pandemic levels, although 2022 still shows some fluctuation as the expanding window adjusts to the structural shock. Most importantly, the seasonal naïve benchmark’s error pattern firmly refutes the presence of seasonality in this market. Although during the 2021 upheaval it experienced large error spikes, like other models, its MASE remains excessively high not only during the shock but also before and after it. This continued poor performance proves that the benchmark incorrectly predicts seasonal patterns, which, in fact, do not exist in the data.

Moreover, an analysis of rank stability across cross-validation folds was conducted to gain a deeper understanding of relative model consistency and whether these findings hold to a greater extent. Dynamic harmonic regression almost always comes out on top for weekly pine log sales, winning the fold (top rank counting) 48% of the time and achieving an average rank of 2.25. However, the pulpwood time series doesn’t show a clear superiority of a single model. The naïve method often achieved the highest rank (first rank in 29% to 32% of folds), but its lead over ETS and LightGBM was so small that it seems model choice is not very important for this product overall. At the quarterly frequency, model rankings become increasingly noisy. Since there were only 45 evaluation folds and the mean performance metrics were nearly tied, the fraction of top-rank finishes is roughly equally divided among the three best models (20% to 27%), making rankings at this level of detail statistically inconclusive. In fact, LightGBM ranks first in 18% to 26% of folds, depending on the configuration, even though it ranks 3rd or 4th on average overall. This shows that the machine learning model can at times beat the statistical models in certain time windows, but it can also perform worse, with errors that are significantly larger in the worst case.

Table 4: Forecasting performance metrics for pine logs and pulpwood across temporal frequencies.

Data Frequency	Model	MASE	RMSE	MAE	MAPE (%)
Pine Logs					
W	DHR	0.704	39.64	33.48	9.35
	ETS	0.750	41.97	35.65	10.09
	LightGBM	0.770	43.08	36.70	10.56
	Naive	0.729	40.79	34.62	9.68
	SeasonalNaive	1.491	81.44	74.00	19.83
M	ARIMA	0.693	37.19	32.29	8.98
	ETS	0.771	41.26	35.80	9.72
	LightGBM	0.775	41.08	35.68	10.10
	Naive	0.726	38.55	33.48	9.36
	SeasonalNaive	1.488	77.84	72.42	19.28
Q	ARIMA	0.968	49.30	45.19	12.66
	ETS	1.038	53.10	48.00	12.04
	LightGBM	0.867	42.99	39.81	11.07
	Naive	0.808	39.59	36.48	10.01
	SeasonalNaive	1.468	73.69	70.24	18.64
Pine Pulpwood					
W	DHR	0.806	26.91	22.57	10.62
	ETS	0.772	25.48	21.39	9.89
	LightGBM	0.879	29.06	24.64	11.70
	Naive	0.816	26.68	22.54	10.46
	SeasonalNaive	1.673	54.84	49.15	21.81
M	ARIMA	0.810	24.02	21.00	9.40
	ETS	0.818	24.20	21.19	9.47
	LightGBM	0.828	24.85	21.76	10.07
	Naive	0.782	23.30	20.33	9.30
	SeasonalNaive	1.714	51.44	47.34	20.79
Q	ARIMA	0.847	22.68	20.79	10.03
	ETS	1.022	28.53	26.92	11.27
	LightGBM	0.840	23.15	21.35	9.70
	Naive	0.812	22.04	20.39	9.15
	SeasonalNaive	1.688	47.73	45.01	19.66

Note: Models are sorted alphabetically within each frequency. The best performing model (lowest MASE) for each frequency and product is highlighted in bold. W, M, and Q represent Weekly, Monthly, and Quarterly frequencies, respectively.

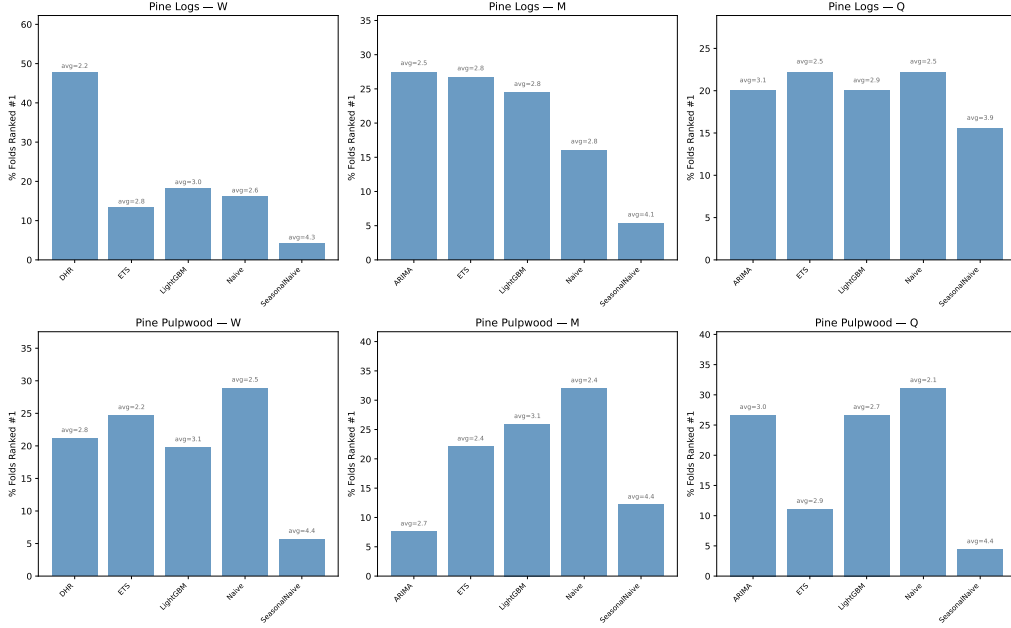
5.1 Performance Degradation During the COVID-19 Structural Break

Between late 2020 and late 2021, Polish timber prices experienced an unprecedented doubling driven by a post-COVID demand recovery, global supply chain disruptions, and rapid housing market growth. This macroeconomic anomaly represents a severe structural break that none of the internal forecasting models used in this paper could predict. To quantify the impact of this shock, we isolate the cross-validation folds covering the price jump and we compare the models' mean absolute scaled error to their results in regular periods, as shown in Figure 3.

The degradation across product types is quite uneven. Pine logs, which are highly influenced by the variability of the construction market, experienced extremely large forecast errors, with error measures increasing from 6.1 to 7.9 times. On the other hand, pine pulpwood showed comparatively less volatility with deterioration from 2.2 to 3.5 times. When the market was so unusual, the difference in forecasting power among the best and worst non-trivial models became almost non-existent. For example, even though DHR usually beats the naïve baseline by about 0.03 MASE under stable market conditions, this edge disappears entirely in the shock scenario. On the other hand, LightGBM seems to degrade the least among the sophisticated models ($\times 6.1$ for logs and $\times 2.9$ for pulpwood). So, we can say that gradient-boosted trees, which have a shorter effective memory, can change price levels a bit more quickly, but this durability is not really enough to make them better than the baseline compared with classic statistical methods. Also worth mentioning is that the size of this single anomaly is so large that it skews the whole paper-wide error averages – not taking the affected folds (Folds 90–100) would bring down all overall mean MASE values by 30–40% and quite possibly change the combined model rankings.

A more thorough analysis of the weekly prediction error patterns during the shock period reveals how

Figure 2: Model rank stability expressed as percentage of folds each model achieves the lowest MASE.



Note: Percentage of expanding cross-validation folds in which each model achieved the highest predictive accuracy (rank 1 based on MASE). The annotated values above the bars denote the model’s average rank across all evaluation folds. W, M, and Q represent Weekly, Monthly, and Quarterly temporal frequencies, respectively.

models recover after a structural break (cf. Appendix A). Prediction errors start increasing rapidly from December 2020 (Fold 91) and reach a maximum just as the market price doubles in April 2021 (Fold 95). This schedule reveals a major difference in the baselines’ recovery paths. Although both the naïve and seasonal naïve models fail disastrously during the rise, the naïve model recovers quickly once the price is set at its new, higher level. By July 2021 (Fold 98), the naïve MASE has fallen to 2.25 as it adjusts its forecasts to the newly established level. On the other hand, the seasonal naïve model’s error keeps getting bigger (MASE = 9.80) since its forecasts are still linked to the pre-shock prices from one year ago. This unbalanced recovery clearly shows why the random walk baseline is inherently more robust for series that undergo permanent structural level changes – it changes immediately, while seasonal autoregressive components are affected by a long-lasting memory of outdated data regimes.

5.2 Cross-Frequency Alignment Results: Weekly to Monthly Aggregation

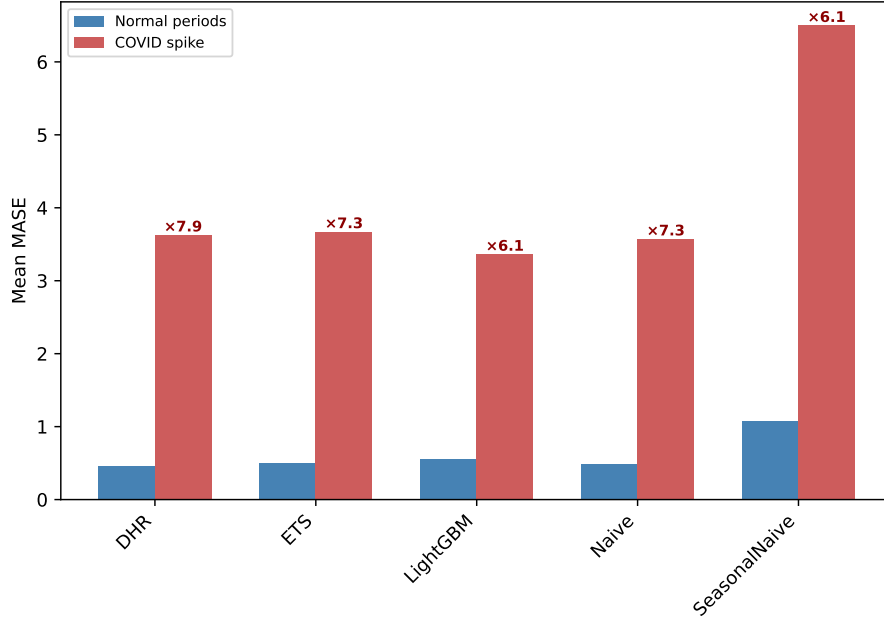
A key operational issue in hierarchical forecasting is whether detailed, high-frequency forecasts, once temporally aggregated, may yield better results than models that are trained directly on lower-frequency data. To do so, we create monthly-resolution forecasts by grouping the 26-week-horizon forecasts from the weekly models into calendar months and computing their arithmetic means. These weekly-aggregated forecasts, which we call $\hat{y}_t^{(W \rightarrow M)}$ are then evaluated against the direct monthly model forecasts, $\hat{y}_t^{(M)}$ through Root Mean Squared Error (RMSE) and the Diebold–Mariano (DM) test for equal predictive accuracy.

The results of the DM tests, summarized in Table 5, indicate a decisive advantage for direct monthly modeling. In 80% of the evaluated model pairs across both products, the direct monthly models demonstrated statistically significant superiority under the squared error loss function ($p < 0.05$). To facilitate a deeper comparison of model’s performance, Table 6 in Appendix B provides a comparison of the RMSE metrics.

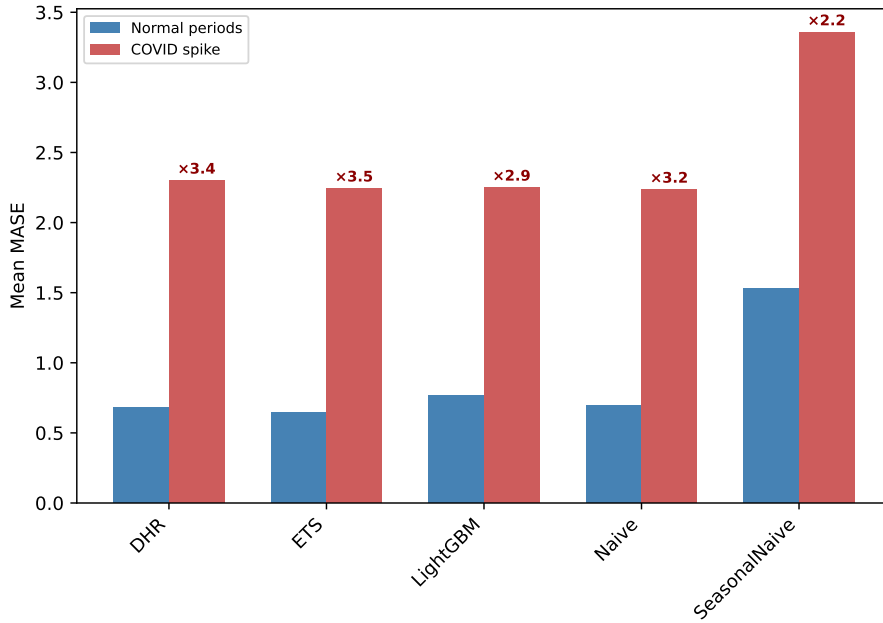
Direct monthly modelling of pine sawlogs generally yielded better results than weekly-to-monthly aggregation, except for one case. When the Dynamic Harmonic Regression (DHR) was aggregated from weekly to monthly, its forecasts were statistically indistinguishable from those of the direct monthly naïve ($p = 0.157$), ARIMA ($p = 0.082$), and LightGBM ($p = 0.654$) models. This indicates that DHR is effectively modeling the true intra-month dynamics, which remain somewhat after temporal aggregation. The best direct monthly model, ETS (RMSE_M = 18.70), on the other hand, was still significantly better than the DHR($W \rightarrow M$) aggregation ($p = 0.004$), highlighting the superiority of direct modeling overall. The pine pulpwood forecasting results are far more decisive. Without exception, all weekly-to-monthly aggregations resulted in significantly larger forecast errors than the best direct monthly models.

The pine pulpwood series, whose dynamics are more stable when modeled weekly, likely end up capturing high-frequency noise rather than an actionable signal, hence the degradation of performance upon

Figure 3: Mean MASE degradation during the 2020–2021 price spike for weekly models.



(a) Pine Logs



(b) Pine Pulpwood

Note: Comparison of model performance during stable market periods versus the COVID-19 structural break. The annotations above the red bars denote the degradation factor (multiplier) of the error during the anomaly.

aggregation.

These alignment findings, from a practical point of view, strongly suggest that forestry organizations and timber market participants choose direct monthly forecasting as their main endeavour. Creating and running weekly models solely for upward aggregation is not only computationally expensive but, as our results show, also inferior in practice for strategic monthly-level decision-making.

5.3 Discussion

A comparative study of the two types of timber assortments reveals that they have different fundamental processes for generating data and different levels of market efficiency. Pine sawlogs have a strong and productive temporal structure that can be exploited compared to pine pulpwood. For sawlogs, the highest predictive capacity was achieved with a weekly-frequency dynamic harmonic regression model, while ARIMA was the most accurate for monthly forecasts. The good performance of these parametric models indicates that sawlog prices exhibit latent autoregressive and pseudo-seasonal structures that are better

Table 5: Diebold-Mariano test summary for Weekly-to-Monthly ($W \rightarrow M$) aggregated forecasts against direct Monthly (M) base models.

Product	Model ($W \rightarrow M$)	Direct Monthly (M) Comparator				
		Naïve(M)	ETS(M)	ARIMA(M)	LightGBM(M)	SeasonalNaïve(M)
Pine logs	Naïve	M^{***}	M^{***}	M^{***}	M^{***}	$W \rightarrow M^{***}$
	SeasonalNaïve	M^{***}	M^{***}	M^{***}	M^{***}	M^{**}
	ETS	M^{***}	M^{***}	M^{***}	M^{***}	$W \rightarrow M^{***}$
	DHR	$n.s.$	M^{***}	$n.s.$	$n.s.$	$W \rightarrow M^{***}$
	LightGBM	M^{***}	M^{***}	M^{***}	M^{***}	$W \rightarrow M^{***}$
Pine pulpwood	Naïve	M^{***}	M^{***}	M^{***}	M^{***}	$W \rightarrow M^{***}$
	SeasonalNaïve	M^{***}	M^{***}	M^{***}	M^{***}	M^{**}
	ETS	M^{**}	M^{***}	M^{***}	$n.s.$	$W \rightarrow M^{***}$
	DHR	M^{***}	M^{***}	M^{***}	M^{***}	$W \rightarrow M^{***}$
	LightGBM	M^{***}	M^{***}	M^{***}	M^{***}	$W \rightarrow M^{***}$

Note: Cells indicate the preferred model based on statistical significance. Statistical significance at the 10%, 5%, and 1% levels is denoted by *, **, and ***, respectively. Lack of statistically significant difference is denoted as $n.s.$.

modeled using sophisticated time-series techniques. On the other hand, pine pulpwood is very close to behaving like a pure random walk. The naïve baseline won against both the monthly and quarterly frequencies for this assortment, while exponential smoothing was the best at the weekly level. This product-specific adaptability aligns directly with previous findings in the Polish market, where ARIMA was identified as the most reliable model for highly volatile assortments, while Winters-Holt (ETS) was highly efficient for stable products (Górna et al., 2022, 2023). Furthermore, the power of the naïve baseline for pulpwood confirms theoretical arguments that, despite potential mean-reverting properties, certain timber price series behave essentially as random walks and are best predicted as such in noisy, limited-sample environments (Khajuria et al., 2012).

The two products also differ radically in their resilience to external shocks, as evidenced by the COVID-19 structural break. At that time, forecasting errors for pine sawlog degraded by a factor of 6.1 to 7.9, while pine pulpwood experienced less fluctuations, with error deterioration going from 2.2 to 3.5 times normal levels. The sawlog market is thus very exposed to macroeconomic demand shocks, whereas the pulpwood market is more insulated. This quantifies a critical vulnerability recently highlighted in the literature, where neither classical nor advanced LSTM models were able to provide precise forecasts of exceptional lumber price increases driven by the COVID-19 housing boom (Verly Lopes et al., 2021; Lamichhane et al., 2023). Furthermore, a random walk baseline showed a greater ability to bounce back from these permanent structural level changes, as it could be adapted immediately to new price levels, whereas models with long autoregressive memory still relied on pre-shock distributions that were outdated. This finding adds empirical weight to the debate on handling structural shifts (Malaty et al., 2007), demonstrating that the naïve model recovers much more quickly than parametric or seasonal models that remain stuck in outdated data regimes.

In addition to examining individual model performance, our results also challenge several methodological assumptions in the forecasting literature. Firstly, the complete failure of the seasonal naïve benchmark shows that for Polish auction prices, isolated predictable yearly seasonality hardly exists and hence cannot be exploited for forecasting. This strongly reaffirms the findings of Banaś and Kożuch (2019), who revealed that seasonal variations in this market are vastly outweighed by cyclic fluctuations and macroeconomic trends. Secondly, in a purely univariate setting, advanced machine learning models such as LightGBM consistently performed worse than statistical models specifically built for the task. This shows that tree-based models typically require macroeconomic exogenous indicators to detect meaningful signals, pointing to an important boundary condition – to surpass the statistical baselines, machine learning models generally require multivariate structures or spatial data (Mei et al., 2010; Thao and Ly Duc, 2025).

6 Conclusions

This study systematically evaluated the forecasting performance of statistical and machine learning models across weekly, monthly, and quarterly frequencies for Polish timber auction prices. Empirical results support the initial hypotheses and offer actionable insights for both time-series modelers and forestry practitioners.

Forecasting in a univariate setting favors simple statistical models over complex tree-based methods. LightGBM consistently underperformed compared with tailored statistical approaches, suggesting that machine learning models typically require exogenous macroeconomic or spatial features to uncover meaningful patterns. It seems hard to ignore that the seasonal naïve method failed completely across all time horizons, indicating no reliable yearly seasonality in timber auction data. Without such signals, seasonal components cannot be used effectively in forecasting efforts.

Testing cross-frequency alignment reveals that combining weekly forecasts to create monthly predictions

is ineffective. Direct monthly modeling consistently delivers better results in Diebold-Mariano comparisons. Higher-frequency data introduces statistical noise that undermines the accuracy of coarser temporal forecasts. The aggregation process does not strengthen predictive power. The evidence supports a clear disadvantage when finer time scales are compressed into broader intervals.

Thirdly, the analysis we carried out on the structural break caused by COVID-19 made us realize that models with long memory of past prices had an important weakness. All models were unable to foresee the exceptional surge in prices. However, the simple naïve baseline model was much better at bouncing back from permanent structural-level changes. The naïve model’s quick adaptation to the newly established price levels makes it recover much faster than parametric or seasonal models, which get stuck in outdated price regimes and data.

For long-term strategic planning or when computational resources are limited, direct monthly modeling with ARIMA or the zero-cost naïve model is often the most reliable and efficient approach. Probably more appropriate for short-term operational procurement are parametric tools such as dynamic harmonic regression for sawlogs and exponential smoothing for pulpwood. It’s empirically suboptimal and statistically harmful to build weekly models solely for upward aggregation in strategic decision-making contexts. Practitioners and forestry organizations should allow their required decision frequency to determine the forecasting method.

Diebold-Mariano tests confirm that direct monthly modeling outperforms combining weekly forecasts and aggregating them upward. The process of stacking finer weekly predictions into monthly outputs adds noise that harms forecasting accuracy. Higher-frequency data contain more variation, which interferes with the stability of lower-frequency estimates. And the statistical evidence shows no advantage in building detailed weekly models for monthly use. Monthly forecasts, when produced directly, are more reliable and consistent under strict evaluation.

Decision frequency in forestry planning must guide model selection. Weekly forecasting for upward aggregation is statistically flawed and empirically inefficient in strategic contexts. For weekly procurement of sawlog or pulpwood, DHR and ETS parametric models are appropriate. At least in theory, monthly ARIMA or naïve models offer greater reliability with minimal computational demand. Direct monthly modeling remains the most efficient strategy for long-term planning or resource-constrained environments. There is no evidence that frequent updating improves forecast accuracy. This approach reduces the risk of overfitting and aligns with observed performance patterns in field data. The naïve model, despite its simplicity, consistently performs well under real-world conditions.

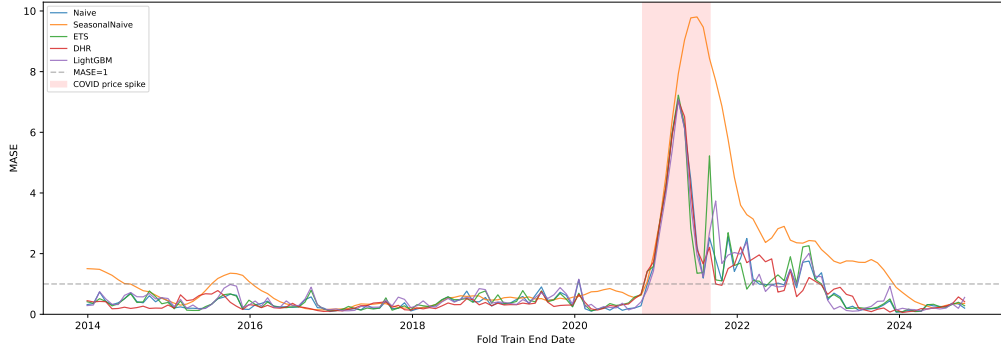
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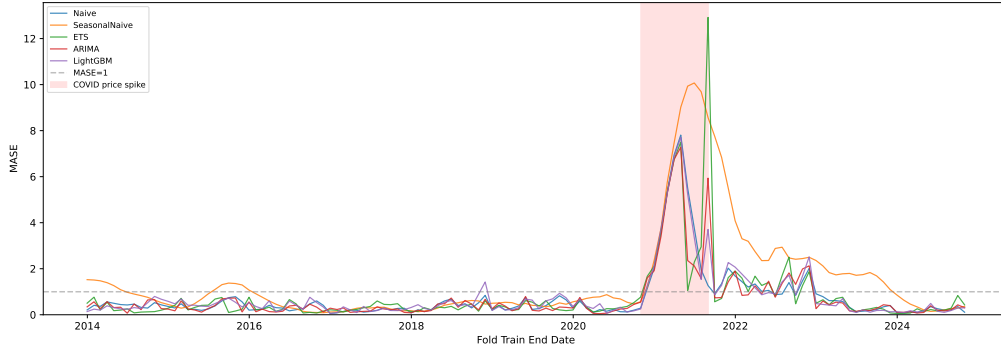
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A Forecasting Performance over Time

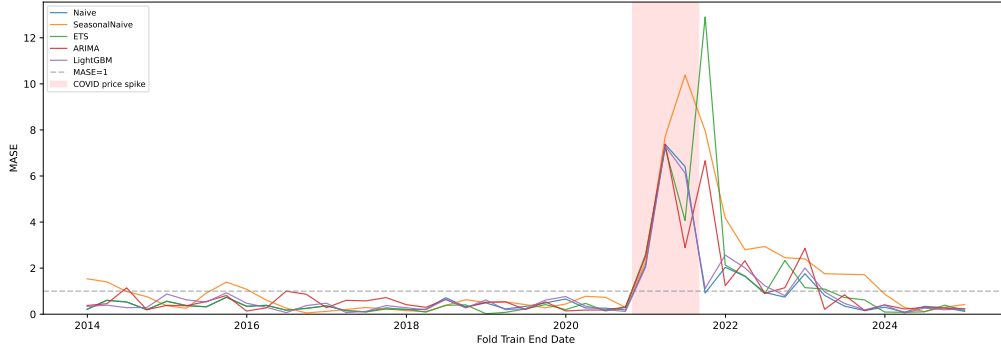
Figure 4: Evolution of Mean Absolute Scaled Error (MASE) over expanding cross-validation folds for Pine Logs.



(a) Weekly Frequency



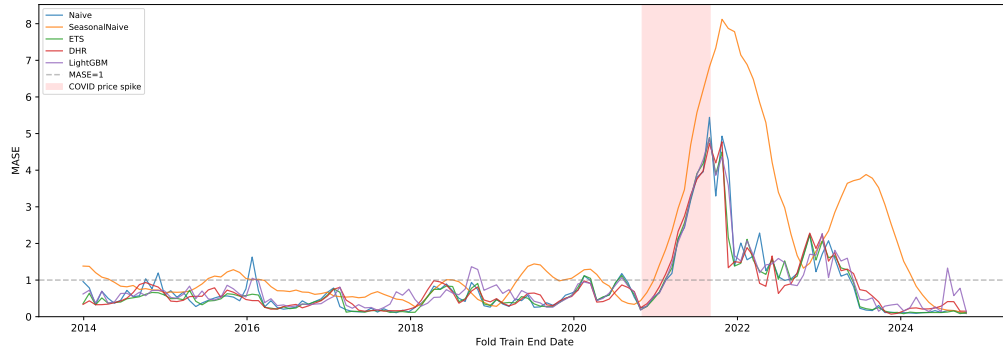
(b) Monthly Frequency



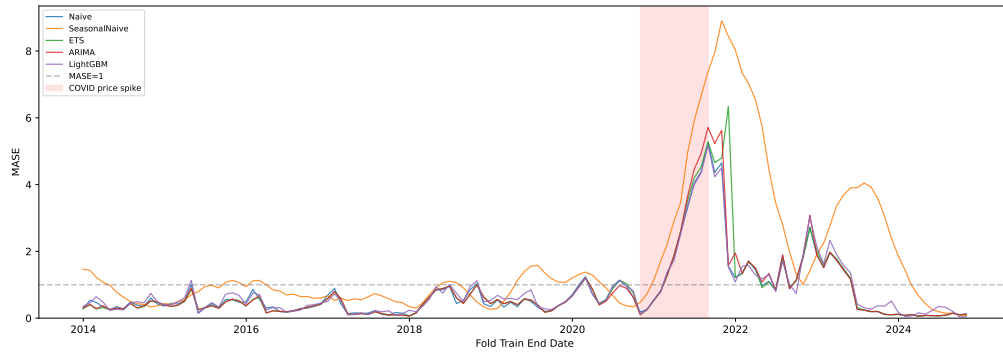
(c) Quarterly Frequency

Note: Expanding window cross-validation MASE trajectories across models for Pine Logs. A MASE value below the dashed line ($y = 1$) indicates performance superior to a seasonal naive benchmark. The shaded red area denotes the COVID-19 related structural price spike.

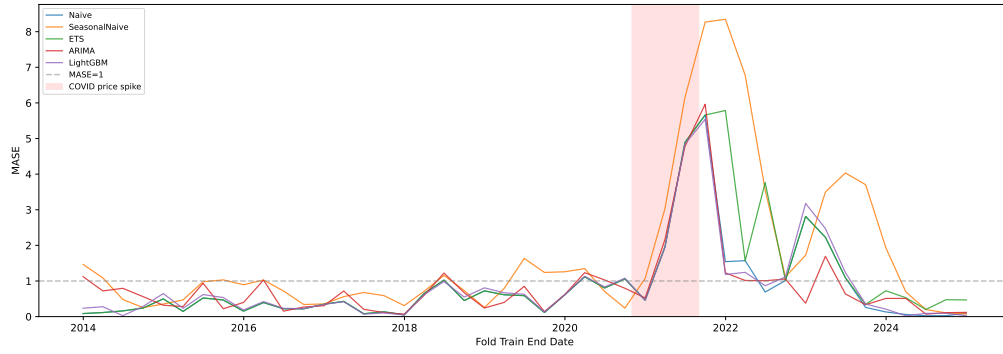
Figure 5: Evolution of Mean Absolute Scaled Error (MASE) over expanding cross-validation folds for Pine Pulpwood.



(a) Weekly Frequency



(b) Monthly Frequency



(c) Quarterly Frequency

Note: Expanding window cross-validation MASE trajectories across models for Pine Pulpwood. A MASE value below the dashed line ($y = 1$) indicates performance superior to a seasonal naïve benchmark. The shaded red area denotes the COVID-19 related structural price spike.

B Detailed Forecast Error Metrics

Table 6: Root Mean Square Error (RMSE) comparison of Weekly-to-Monthly ($W \rightarrow M$) aggregated forecasts versus direct Monthly (M) base forecasts.

Product	Model Family	RMSE ($W \rightarrow M$)	RMSE (Direct M)	Ratio ($\frac{W \rightarrow M}{M}$)
Pine logs	Naive	25.20	20.45	1.23
	Seasonal Naive	43.54	42.60	1.02
	ETS	26.05	18.70	1.39
	DHR (W) / ARIMA (M) [†]	22.34	19.98	1.12
	LightGBM	31.38	21.72	1.44
Pine pulpwood	Naive	16.71	11.61	1.44
	Seasonal Naive	22.64	21.85	1.04
	ETS	13.33	10.74	1.24
	DHR (W) / ARIMA (M) [†]	16.27	10.91	1.49
	LightGBM	17.40	12.56	1.39

Note: Models are grouped by their methodological family. [†] For the weekly frequency, Dynamic Harmonic Regression (DHR) was utilized, whereas ARIMA was utilized for the direct monthly frequency. The ratio $\text{RMSE}_{W \rightarrow M} / \text{RMSE}_M$ illustrates the magnitude of the accuracy loss (or gain) incurred by aggregating high-frequency forecasts rather than modeling the target frequency directly. A ratio greater than 1.0 indicates that the direct monthly model achieved a lower forecasting error.